

Assignment #4

Stat 332: Sampling and Experimental Design (Winter 2026)

Professor: Christian Boudreau

Due at 11:59 pm on Monday Apr 6, 2026

Instructions:

- Teamwork is allowed and encouraged; however, everyone must hand in their own assignment showing that they understood what they wrote. You are not allowed to collaborate with anyone who is not a STAT 332 classmate.
- Justify all your answers.
- One solution per question; if you provide multiple solutions, you will receive $\frac{1}{2}$ the points of the inferior solution.
- Write legibly; if the TA or I have to guess at your answers, it's unlikely to be in your favour.
- Make sure to include your R code and relevant output for Q#2 and Q#5.
- Unless specified otherwise, you are allowed to use R to perform any calculations. However, you are not allowed to use any functions of the [R survey package](#) unless instructed to do so in the question.
- Crowdmark:
 - You should soon received an automated email from Crowdmark saying that you can now submit/upload your solution. Note that, even if you did not get the email, you should still be able to submit/upload your assignment by simply going to [Crowdmark](#), and signing into your account (via UW Learn).
 - To ensure smooth and efficient grading, please make sure to upload your solution separately for each question.
 - Crowdmark allows you to resubmit anytime before the due date; **once the due date has passed, your submission is locked in and you will thus not be able to resubmit.**
 - Late submissions will be accepted, but are subject to a 5% per hour penalty*; click [here](#) for more information on Crowdmark and late submissions.
 - Technical difficulties with Crowdmark:
 - Consult [Crowdmark Help](#)
 - Watch [this short video](#) about submitting an assignment on Crowdmark
 - [E-mail me](#)
 - **If tragedy strikes and you cannot upload your assignment to Crowdmark, email it to me**, so I have proof of when you completed your assignment (otherwise, the above mentioned lateness penalty will apply).
- Recall that each of you is allowed a one-time extension with a reduced lateness penalty. You have 3 option: 1) a 5% reduced penalty for an additional 24 hours, 2) a 10% reduced penalty for an additional 48 hours, and 3) a 0% reduced lateness penalty if you submit your assignment within 3 hours past the deadline.

To take advantage of the first two options (i.e., 5% and 10%), you have to [email me](#) (at least) 6 hours before the original due date. In both cases, I will extend your due date on Crowdmark by 48 hours. I you submit within 24 hours, you will automatically get the 5% reduced penalty; if not, you will automatically be switched to the 10% reduced penalty. Please note that if you do not submit your assignment within the 48 hour extension, you will incur both the reduced lateness penalty (i.e., 10%) plus the standard penalty of 5% per hour.

Since this is meant to be a last minute thing, you do not need to email me in advance to take advantage of the 0% (3 hours) lateness penalty. However, since it counts towards your one-time reduced lateness penalty, you still need to [email me](#) (within 24 hours of the original due date) to inform me that you want to take advantage of this option.
- See [course outline](#) for more information on assignments, and how much each of them contributes to your overall grade.

*This is the default penalty, and is applied when there are no extenuating circumstances.

Question #1 (10 points):

This question uses the pedestrian crossing button of assignment #3 to get a better understanding of how contrasts work. Recall from the assignment that an experiment consisting of a grand total of 32 observations was conducted to investigate the waiting time (in seconds) for the pedestrian crossing signal to turn white after pushing the button 0, 1, 2 and 3 times. Please consult assignment #3 for the data.

- (a) Give a contrast to compare the effect of no pushes vs. pushing the button once or more. Estimate that contrast, and give the corresponding 95% CI.
No need to make multiple comparison adjustment, as this is the only contrast of interest.
- (b) Multiply the c_i 's of the contrast in part (a) by 3, and compute the corresponding 95% CI.
Again, no need to make multiple comparison adjustment.
- (c) Are you surprised that parts (a) and (b) do not yield the same CI? Briefly explain.
- (d) The fact that parts (a) and (b) do not yield the same answer is annoying since (ultimately) both set of contrasts do correctly test for the effect of no pushes vs. pushing the button once or more. However, what happens when we are doing the actual test instead of using a CI? It can be shown (you do not need to prove this) that the test statistics for confronting $H_0 : C = 0$ vs $H_1 : C \neq 0$ is given by

$$t_{\text{obs}}^* = \frac{\hat{C}}{\sqrt{\hat{V}(\hat{C})}}$$

The corresponding t -test consists in rejecting H_0 (in favour of H_1) if $|t_{\text{obs}}^*| > t_{n-I; 1-\alpha/2}$.

- d.i) Compute t_{obs}^* corresponding to part (a), and perform the t -test. Do you reject H_0 ?
Recall that $t_{28; 0.975} = 2.048$.
- d.ii) Compute t_{obs}^* corresponding to part (b), and perform the t -test. Do you reject H_0 ?
- d.iii) Compare d.i and d.ii; do you get the same result(s)?
- d.iv) Clearly explain why multiplying by a constant doesn't impact the t -test.

Question #2 (10 points):

I made a small experiment to see how long it takes to boil (in minutes) a given amount of water on the four different burners of my stove (1=right-back burner, 2=right-front, 3=left-back and 4=left-front). Since salt affects the boiling point of water, I ran the experiment with 0, 2, 4 and 6 teaspoons of salt added to the water. Hence, my experiment has two factors (i.e., burner and salt, with 4 levels each). In addition, I repeated the experiment 3 times for each of the 16 burner/salt combinations; yielding a total of 48 observations. The corresponding dataset is available on the [STAT 332 LEARN website](#); see file `water.txt`.

Use **R** to answer the following:

- (a) Produce the ANOVA table.
- (b) Perform the usual F tests in a two-way ANOVA. What are your conclusions on the effects of salt, burner and their interaction?
- (c) Using pairwise comparisons, which burner or burners is/are the fastest at boiling water? (Make sure that you clearly justify why only this/those burner/s is/are faster than the others.)
- (d) Use a contrast to test (at the 5% level) if the two right burners take the same time to boil water than the left ones. *No need to make multiple comparison adjustment.*
- (e) Use a contrast to test (at the 5% level) the hypothesis that there is a linear trend in the time to boil water due to the level of salt. *Again, no need to make multiple comparison adjustment.*

Question #3 (10 points):

- (a) In a two-way ANOVA, let S_{ij}^2 be the sample variance of the n_0 observations of the (i, j) treatment (i.e.; i^{th} level of factor A and j^{th} level of factor B); in other words,

$$S_{ij}^2 = \frac{1}{n_0 - 1} \sum_{k=1}^{n_0} (y_{ijk} - \bar{y}_{ij\bullet})^2$$

Prove that S_{ij}^2 is an unbiased estimator of σ^2 .

- (b) When the F test for interactions is not statistically significant, an alternate estimator of σ^2 is obtained by pooling SSE and SSAB. Let $\text{SSP} = \text{SSE} + \text{SSAB}$, and

$$\text{MSP} = \frac{\text{SSP}}{n - I - J + 1}$$

be the corresponding mean squares. Show that MSP is an unbiased estimator of σ^2 when there are no interactions.

Question #4 (10 points):

In tutorial #7 we proved that in a two-way Anova, $E(\text{MSE}) = \sigma^2$ and $E(\text{MSA}) = \sigma^2 + n_0 J / (I - 1) \sum_{i=1}^I \alpha_i^2$. This question builds on that.

(a) Show that in a randomized block design model

$$E(\text{MSBL}) = \sigma^2 + \frac{I}{R-1} \sum_{j=1}^R b_j^2$$

where $\text{MSBL} = \text{SSBL} / (R - 1)$ and

$$\text{SSBL} = I \sum_{j=1}^R (\bar{y}_{\bullet j} - \bar{y}_{\bullet\bullet})^2$$

(b) Part (a) is (obviously) for the randomized block design model of chapter 9; i.e.,

$$Y_{ij} = \mu + \alpha_i + b_j + \varepsilon_{ij}$$

where $i = 1, \dots, I$, $j = 1, \dots, R$ and ε_{ij} i.i.d. $N(0, \sigma^2)$. However, it is also possible to view/model the b_j 's as random. Doing so yields the same model as above with the addition that b_j i.i.d. $N(0, \theta)$ and that the ε_{ij} 's are independent of the b_j 's. This means that we no longer need that $\sum b_j = 0$. It also means that $V(Y_{ij}) = \sigma^2 + \theta$ and that $\text{Cov}(Y_{ij}, Y_{kj}) = \theta$ for $i \neq k$.

Show that under such a model,

$$E(\text{MSBL}) = \sigma^2 + I\theta$$

Recall that we can no longer assume that $\sum b_j = 0$.

Question #5 (10 points):

This question is a follow-up to Q#4 of Assignment #3, and is thus also concerned with making paper frogs. My daughter and I did a more thorough experiment, and investigated both the effects for dimensions (i.e., $10\text{cm} \times 10\text{cm}$ vs $15\text{cm} \times 15\text{cm}$ vs $20\text{cm} \times 20\text{cm}$) and colour (i.e., gray vs orchid) on the jumping distance of paper frogs. After making the 6 paper frogs ($3 \text{ dimensions} \times 2 \text{ colours}$) and practicing a few times, we individually launch each frog 5 times; for a grand total of 60 jumps (and thus 60 measurements). Our data is available on the [STAT 332 LEARN website](#); see file `frogs.xlsx`. This file can be easily read into R by using the `read_excel` function (which is best used by in combination with the `as.data.frame` function) of the `readxl` package, and contains the following variables:

Variable	Description
<code>dimensions</code>	Paper dimensions ($10\text{cm} \times 10\text{cm}$ vs $15\text{cm} \times 15\text{cm}$ vs $20\text{cm} \times 20\text{cm}$)
<code>colour</code>	Paper colour (gray vs orchid)
<code>jump</code>	How far the frog jumped (in cm)
<code>person</code>	Who launched the paper frog (Christian/me vs my daughter)

Use R to answer the following:

- (a) Fit a two-way Anova model where the two factors are paper dimension (variable `dimensions`) and colour (variable `colour`), and answer the following:
Note: this model simply ignores the person who launched the paper frog.
- a.i) Produce the two-way Anova table, and perform the usual F tests. What are your conclusions on the effects of paper dimensions, colour and their interaction?
 - a.ii) Compute all relevant pairwise comparisons and, using Tukey's method, obtain their CI's with a family level of 95%.
 - a.iii) Using part a.ii), determine which one (or ones) of groups jumps the furthest?
- (b) Expand on the two-way Anova model fitted in part (a) by including `person` as a blocking factor. In other words, fit a two-way Anova model with `dimensions` and `colour` as factors (don't forget to include their interaction), and `person` as a blocking factor.
- b.i) Produce the two-way Anova table, and perform the usual F tests. What are your conclusions on the effects of paper dimensions, colour and their interaction? Are they the same as in a.i)?
 - b.ii) In section 9.6 of the slides, I mentioned that estimating and comparing the blocks is usually not of interest. Though this is true, for this particular problem, my daughter nevertheless want to know if I she is better than I at making the paper frogs jump further. Compute a 95% CI to test this?
 - b.iii) Repeat a.ii and a.iii using the two-way Anova model with blocking. Do you reach the same conclusion about which one (or ones) of groups jumps the furthest?
 - b.iv) Check the assumptions of the Anova model, and look for possible outliers. In particular, test the normality assumption, 0 expectation of errors and equality of variance.